WORKED SOLUTION

STEP II 2021, Question 4

Pure • Lambert W Function

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Read the question before you go on. It is in the STEP II paper in OCR's download of the 2021 STEP papers (a zip file):

www.ocr.org.uk/Images/696588-2021-past-papers.zip

This question has a full guided treatment in *Beyond STEP*, Section 3.9, *Inventing a New Function*: https://beyondstep.keithmcnulty.org/ch03/lambert-w.html

The key idea

The function g undoes xe^x, so the whole question is about rearranging each equation into the shape $u\,e^u = c$ for some new variable u. Once it is in that shape, the sketch from part (i) tells us everything: there is no solution if $c < -\frac{1}{e}$, and two solutions if $-\frac{1}{e} < c < 0$, of which g only ever picks out the one with $u \geq -1$. Logarithms are dealt with by substituting $x = e^u$.

Solution

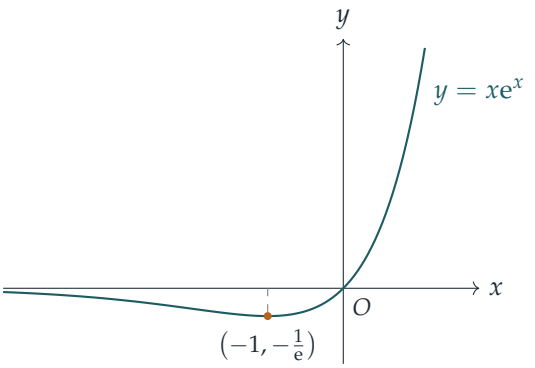
Part (i)

Let $y = xe^x$. By the product rule,

$$\frac{dy}{dx} = (1+x)e^x,$$

which is zero only at $x = -1$, since $e^x > 0$. The gradient is negative for $x < -1$ and positive for $x > -1$, so the only stationary point is a minimum at $\left(-1, -\frac{1}{e}\right)$.

The curve has the sign of x and passes through the origin. As $x \to \infty$, $y \to \infty$ rapidly. As $x \to -\infty$, e^x beats x, so $y \to 0$ from below: the negative x-axis is an asymptote.



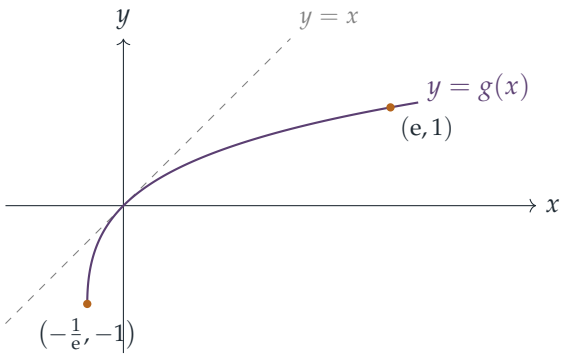
Part (ii)

For f to have an inverse it must be one-to-one on $[a, \infty)$. If $a < -1$, the interval contains points on both sides of the minimum near $x = -1$, and since f is continuous and strictly decreasing on $(-\infty, -1]$ and strictly increasing on $[-1, \infty)$, some value just above $-\frac{1}{\mathrm{e}}$ is taken twice. If $a \geq -1$, f is strictly increasing, hence one-to-one. So the minimum possible value is

$$a = -1.$$

Then f maps $[-1, \infty)$ onto $[-\frac{1}{\mathrm{e}}, \infty)$, so g has domain $[-\frac{1}{\mathrm{e}}, \infty)$ and range $[-1, \infty)$.

The graph of g is the reflection of the right-hand branch of part (i) in the line $y = x$. It starts at $(-\frac{1}{\mathrm{e}}, -1)$ with a vertical tangent (the reflection of the horizontal tangent at the minimum), passes through the origin, where it has gradient 1 (as $f'(0) = 1$), and increases ever more slowly: $g(\mathrm{e}) = 1$, and $g(x)$ grows more slowly than $\ln x$ for large x.



Two facts will be used repeatedly in part (iii). For $c \geq -\frac{1}{\mathrm{e}}$, $u = g(c)$ is the solution of $u\mathrm{e}^u = c$ with $u \geq -1$. And from the sketch in part (i), the equation $u\mathrm{e}^u = c$ has

- no real solution if $c < -\frac{1}{\mathrm{e}}$;
- two real solutions if $-\frac{1}{\mathrm{e}} < c < 0$, namely $g(c) \in (-1, 0)$ and another less than -1;
- exactly one real solution, $g(c)$, if $c \geq 0$ (for $u < 0$ the left-hand side is negative).

Part (iii)

(a) Multiplying by e^x (which is never zero) gives

$$xe^x = \tfrac{1}{5}.$$

Since $\frac{1}{5} > 0$, there is exactly one real root:

$$x = g(\tfrac{1}{5}).$$

(b) We need $x > 0$ for $\ln x$ to exist, so put $x = e^u$ with $u = \ln x$ real. The equation becomes

$$2ue^u + 1 = 0 \iff ue^u = -\tfrac{1}{2}.$$

But $ue^u \geq -\frac{1}{e}$ for all u, and $-\frac{1}{2} < -\frac{1}{e}$ because $e > 2$. So there is *no real root*.

(c) With $x = e^u$ as before,

$$3ue^u + 1 = 0 \iff ue^u = -\tfrac{1}{3}.$$

Now $-\frac{1}{e} < -\frac{1}{3} < 0$ because $e < 3$, so there are two values of u, and hence (as $u \mapsto e^u$ is one-to-one) two real roots x. One of them is

$$x = e^{g(-1/3)}.$$

? **Ask yourself** Is this the larger or the smaller root?

The other value of u is less than -1, while $g(-\frac{1}{3}) > -1$. Since e^u is increasing in u, the root $e^{g(-1/3)}$ is the *greater* of the two roots.

(d) Again $x > 0$. Exponentiating, $x = 3\ln x \iff e^{x/3} = x \iff xe^{-x/3} = 1$. To get the form ue^u we need the multiplier in front to match the exponent, so put $u = -\frac{x}{3}$ and multiply by $-\frac{1}{3}$:

$$\left(-\frac{x}{3}\right)e^{-x/3} = -\frac{1}{3}.$$

As in (c), $-\frac{1}{e} < -\frac{1}{3} < 0$, so there are two values of u, both negative, giving two positive roots $x = -3u$. One root is

$$x = -3\,g(-\tfrac{1}{3}).$$

This time the map $u \mapsto x = -3u$ is *decreasing*. The other value of u is less than -1, so the other root is greater than 3, while our root satisfies $x = -3g(-\frac{1}{3}) < 3$. So the root we have found is the *smaller* of the two.

Part (iv)

Since $x > 0$, take logarithms: $x^x = 10 \iff x\ln x = \ln 10$. Put $x = e^u$:

$$ue^u = \ln 10.$$

As $\ln 10 > 0$, this has the single solution $u = g(\ln 10)$, so the root is

$$x = e^{g(\ln 10)}.$$

(Equivalently, since $e^{g(c)} = c/g(c)$ for $c \neq 0$, the root is $\dfrac{\ln 10}{g(\ln 10)}$.)

Checks

- Numerically, $g(-\frac{1}{3}) \approx -0.6191$. In (c) this gives $x \approx 0.5384$, and $3(0.5384)\ln(0.5384) \approx -1.000$; the other root is about 0.2204, which is indeed smaller.
- In (d), $x \approx 1.8572$, and $3\ln 1.8572 \approx 1.8572$; the other root is about $4.536 > 3$, as predicted.
- In (iv), $g(\ln 10) \approx 0.9188$, so $x \approx 2.5062$, and $2.5062^{2.5062} \approx 10.00$.
- In (b), the smallest value of $2x\ln x + 1$ is $1 - \frac{2}{\mathrm{e}} \approx 0.264 > 0$ (at $x = \frac{1}{\mathrm{e}}$), confirming there is no root.

Common slips

- Taking $a = 0$ in part (ii) because the curve "looks" one-to-one there: the turning point is at $x = -1$, not at the origin.
- In (d), forgetting to multiply through by $-\frac{1}{3}$, so that the factor in front does not match the exponent and g cannot be applied.
- Assuming the root found via g is always the larger one: it is in (c), where $x = \mathrm{e}^u$ is increasing in u, but not in (d), where $x = -3u$ reverses the order.

Beyond STEP

The function g is the principal branch of the *Lambert W function*, usually written W_0; the other solution of $u\mathrm{e}^u = c$ for $-\frac{1}{\mathrm{e}} < c < 0$ comes from a second branch, W_{-1}. It turns up in delay differential equations, in combinatorics (counting labelled trees) and in physics, wherever an unknown appears both inside and outside an exponential. Continuing an inverse function across the point where it fails to be one-to-one is the start of the theory of branches and branch points in complex analysis; Needham's *Visual Complex Analysis* is a beautiful place to meet these ideas.

Beyond STEP worked solutions: https://beyondstep.keithmcnulty.org/. Solution text CC0 1.0; the question itself is © Cambridge University Press & Assessment and is not reproduced here.